UNIVERSITI TEKNOLOGI MALAYSIA

Faculty of Computing

SECP3623

Database Programming

2025/2026 SEMESTER 1

Assignment 2 (10%) – Transaction Management

Name	:	
Matric Id	:	
Section	:	

Instructions:

This assignment evaluates your ability to apply the concept of serializability, differentiate between serial and non-serial transaction schedules and implement locking techniques to manage simultaneous transactions.

Academic Integrity

All students must work in a group of **TWO(2)**. The use of AI-generated tools or sharing of answers between students **is strictly prohibited** and will result in **zero marks**. You are required to complete the answers in **handwritten form**. Please take **clear screenshots** of your answers and submit them in **PDF format**.

QUESTION 1 (15 MARKS)

Consider the schedule transactions in Table 1 and answer all following questions (i) – (iv).

Table 1: Schedule T1, T2 and T3

Time	T1	T2	T3
t_1	begin transaction		
t_2	read_item(X)		
t_3	read_item(Y)	begin transaction	
t_4	commit	read_item(M)	begin transaction
t_5		read_item(N)	read_item(N)
t_6		M=(N+M)*10	read_item(M)
t_7		write_item(M)	N=N+150
t_8		commit	write_item(N)
t_9			commit

i) Draw the precedence graph for the transaction schedule in Table 1 to check for serializability.

ii) Explain the precedence graph produced in (i).

iii) Apply the locking technique to the transaction schedule in Table 1, and revise the transaction schedule in Table 1 using the following format:

T1	T2	T3

iv) Identify the problem(s) that arise in (iii) and then explain any potential problems that may still occur after applying the locking technique in concurrency control based on your answer in (iii).

QUESTION 2 (15 MARKS)

Based on the transaction schedule in Table 2, answer all the following questions:

Table 2: Transaction Schedule for T4-T5-T6

	T4	**T5**	**T6**
	begin transaction		
	read_item(P)		
	read_item(Q)	begin transaction	
↓	P=(P*5)+Q	read_item(Q)	
time	write_item(P)	read_item(R)	begin transaction
	commit	Q=Q+R	read_item(R)
		write_item(Q)	R=R-100
		commit	write_item(R)
			commit

i) Draw the precedence graph for the transaction schedule in Table 2 to check for serializability.

ii) Explain the precedence graph produced in (i).

iii) Apply the two-phase locking (2PL) protocol to emphasize the positioning of lock and unlock operations in each transaction using the following format.

T4	T5	T6

QUESTION 3 (25 MARKS)

a) Consider the schedule transactions in Table 3.1 and answer all following questions (i) – (iv).

Table 3.1: Schedule T7, T8, T9 and T10

Time	**T7**	**T8**	**T9**	**T10**
t_1	begin transaction	begin transaction		
t_2	read (X)	read (Y)	begin transaction	
t_3	X = X + 50		read (X)	begin transaction
t_4	write (X)			read (Z)
t_5		read (X)	X = X - 15	Z = Z + 20
t_6		X = Y * 12	write (X)	write (Z)
t_7		write (X)	commit	
t_8	commit	commit		commit

i) Is the schedule in Table 3.1 a serial or non-serial schedule? State your reason.

ii) Draw the precedence graph for the transactions given in Table 3.1.

iii) Does the schedule in Table 3.1 have conflict-serializable or not? How can you determine this?

iv) What problem does the schedule in Table 3.1 face?

b) Consider the schedule transactions in Table 3.2 and answer all following questions (i) – (iv).

Table 3.2: Schedule T11, T12, T13 and T14

Time	T11	T12	T13	T14
t_1	begin transaction			
t_2	read (A)	begin transaction		
t_3		read (B)	begin transaction	
t_4	A = A + 200	B = B - 50	read (A)	begin transaction
t_5	write (A)			read (C)
t_6		write (B)	A = A - 30	C = C + 10
t_7			write (A)	write (C)
t_8	commit			
t_9		read (A)		
t_{10}		A = A * 2		
t_{11}		write (A)		
t_{12}		commit	commit	commit

i) Implement the basic locking to the schedule in Table 3.2 by specifying the appropriate lock type (shared or exclusive) for each transaction.

ii) Based on your answer in (i), state any other issues that may arise.

iii) Implement the Two-Phase Locking (2PL) protocol to the schedule in Table 3.2.

iv) Based on your answer in (iii), state any other issues that may arise.

QUESTION 4 (15 MARKS)

Consider the schedule transactions in Table 4 and answer all following questions (i) – (v).

Table 4.1: Schedule T15, T16 and T17

Time	T15	T16	T17
t_1	begin transaction		
t_2	read (J)	begin transaction	
t_3	J = J + 10	read (K)	begin transaction
t_4	write (J)	K = K - 5	read (J)
t_5	Read (K)	write (K)	J = J - 15
t_6	K = K + 20		write (J)
t_7	write (K)	commit	
t_8	commit		commit

i) Is the schedule in Table 4 a serial or non-serial schedule? State your reason.
ii) Draw the precedence graph for the transactions given in Table 4.
iii) Does the schedule in Table 4 have conflict-serializable or not? How can you determine this?
iv) What problem does the schedule in Table 4 face?
v) Apply the two-phase locking (2PL) protocol to emphasize the positioning of lock and unlock operations in each transaction using the following format.

Submission

1. Submission due date: **24 December 2025 (before 10:00PM)**.
2. Each group must submit only **ONE** answer in PDF format.
3. All work must be original and completed independently. The use of AI-generated tools or sharing of answers between students is strictly prohibited.
4. Late submissions will be penalized according to faculty policy